# From chat to measured results.

## Discover. Evaluate. Choose with evidence.

ModelPicker turns a Claude conversation into a reproducible CPU benchmark of text classifiers. Discover Hugging Face models, import labeled data, compare quality and speed, and export results. Supports single-label sentiment, spam and ticket classification.

| 23 | 5 | 1,000 | CPU |
|---|---|---|---|
| MCP tools | models per plan | rows per model | local inference |

### Discover & import

Search models and datasets; inspect labels and revisions. Import CSV, JSONL, JSON, pasted rows, trusted URLs or Hub datasets.

### Plan & evaluate

Seeded sampling, pinned revisions and identical rows. Background jobs with progress, cancellation and resource limits.

### Measure & reuse

Accuracy, macro F1, per-label scores, latency, throughput and memory. Save runs; export reports and Python integration code.

Python · MCP / FastMCP · PyTorch · Transformers · Hugging Face · SQLite

### 01 / LOCAL CONNECTION

1. In the project environment, run `python -m modelpicker setup`.
2. Merge `~/.modelpicker/desktop-config.json` → `mcpServers.modelpicker` into Claude's config: Desktop app → Local MCP servers → Edit config.
3. Fully quit and reopen Claude Desktop; enable ModelPicker in the chat tools menu.

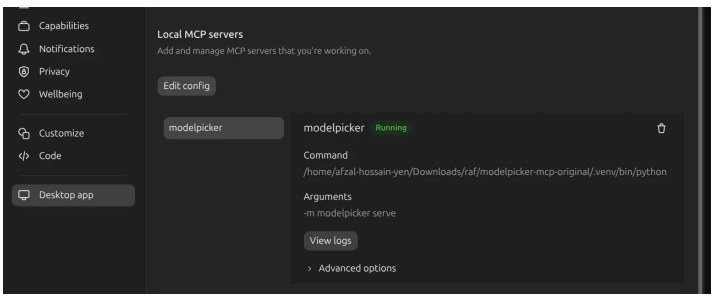
**stdio · No ModelPicker OAuth · Uses this computer's CPU**

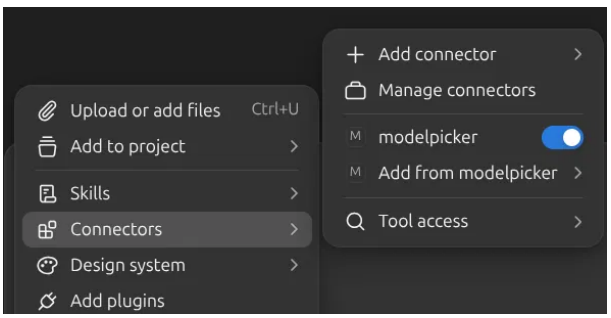
### 02 / REMOTE CONNECTION

1. Run setup, then create a public HTTPS tunnel:
`cloudflared tunnel --url http://127.0.0.1:7860`
2. Use the tunnel's HTTPS origin to start the server (one command):
`python -m modelpicker serve --transport streamable-http --public-url https://YOUR-TUNNEL-HOST`
3. Claude Settings → Connectors → Add custom connector:
`https://YOUR-TUNNEL-HOST/mcp`. Authorize in the browser using `~/.modelpicker/owner-password.txt`. Keep the server and tunnel running.

Uses the server host's CPU. Quick-tunnel URLs change on restart; reconnect after server restarts. One trusted owner per installation. Use one HTTP service for multiple clients; avoid concurrent servers sharing storage.

### CONNECTED IN CLAUDE DESKTOP



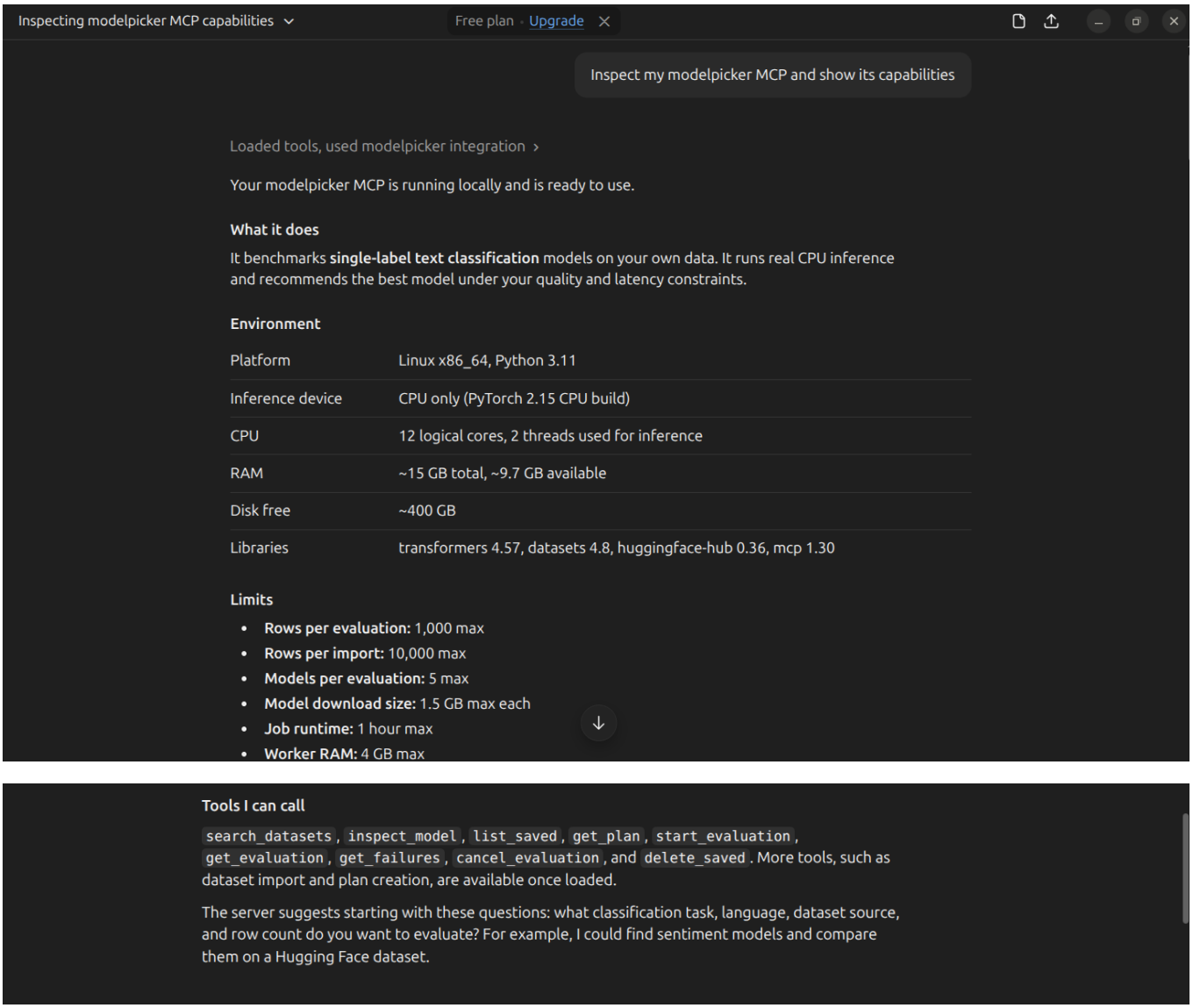

Text and tool results sent to Claude reach the provider. No paid inference API is required; downloads and hardware costs still apply.

# 01 / Inspect capabilities

### “Inspect my modelpicker MCP and show its capabilities”

Implemented workflow tools: get_capabilities

View conversation



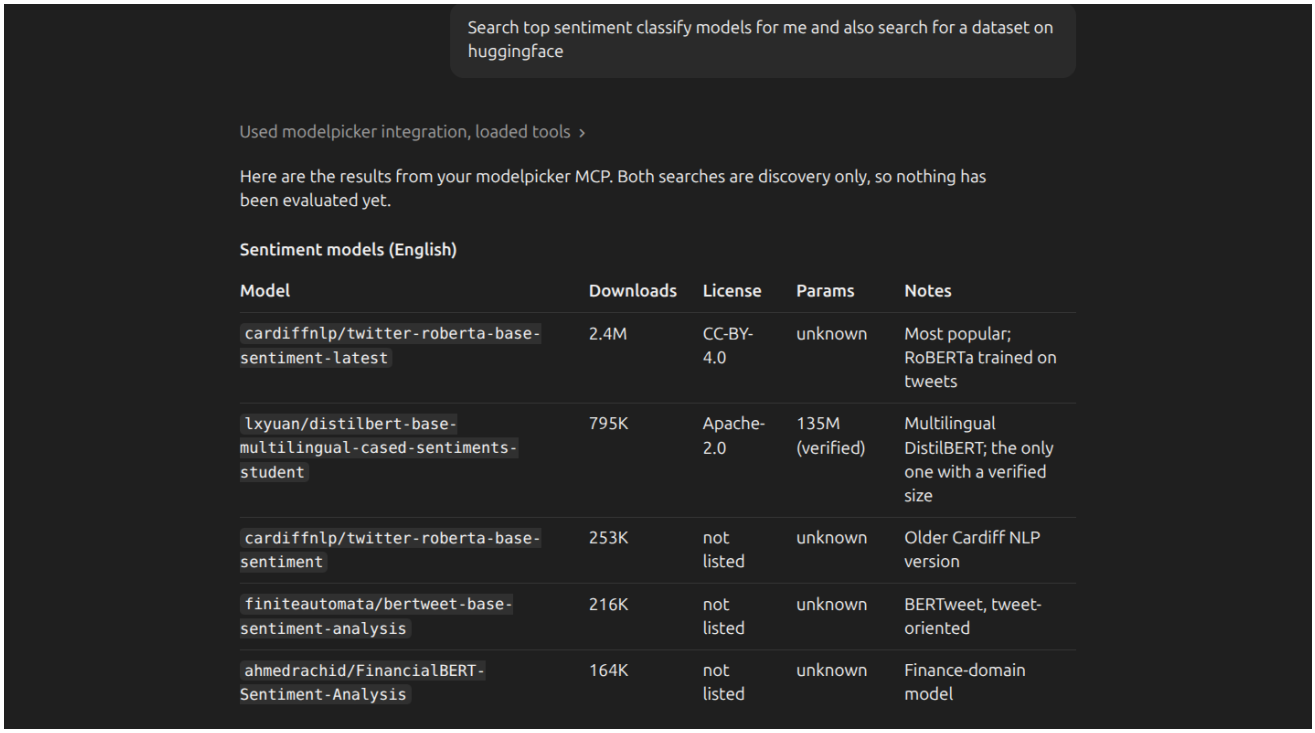

The session reports the local CPU environment, package versions and resource limits. These are historical observations from this session. Every tool name visible in the screenshot exists in the project.

# 02 / Discover models & datasets

### “Search top sentiment classify models for me and also search for a dataset on huggingface”

Implemented workflow tools: search_models · search_datasets

View conversation



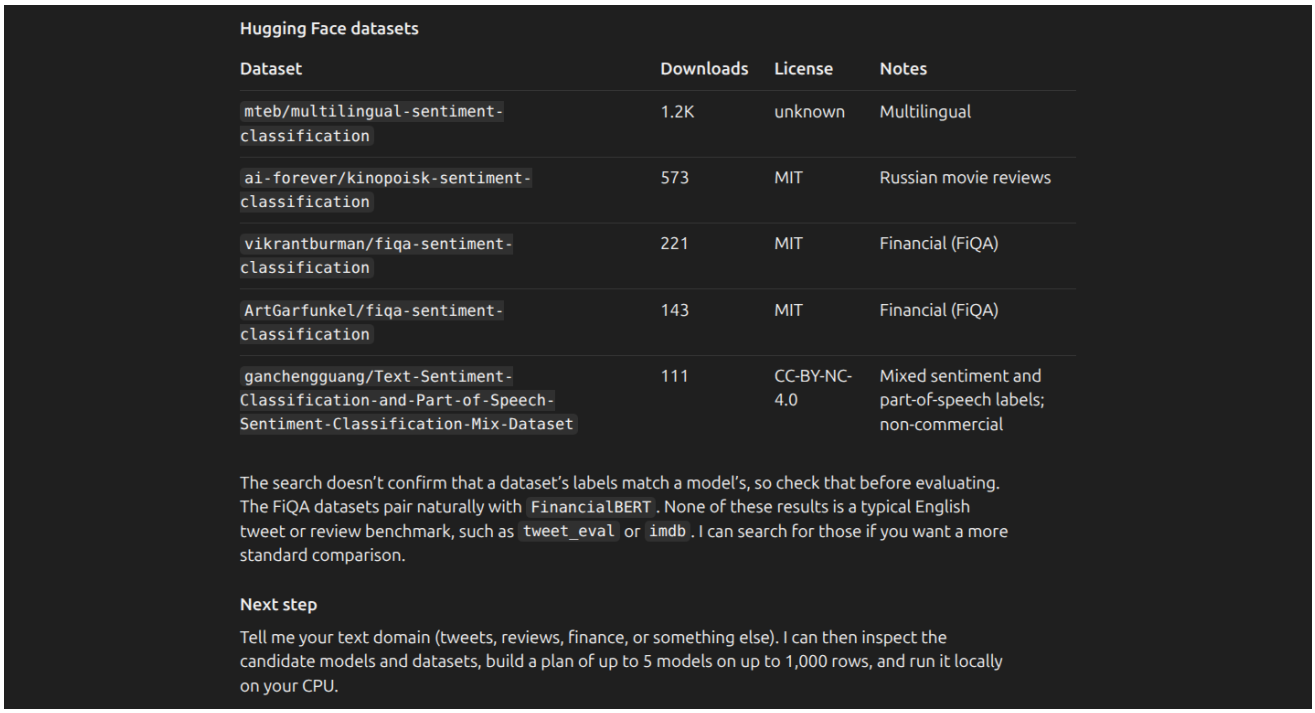
Search top sentiment classify models for me and also search for a dataset on huggingface

Used modelpicker integration, loaded tools ›

Here are the results from your modelpicker MCP. Both searches are discovery only, so nothing has been evaluated yet.

**Sentiment models (English)**

| Model | Downloads | License | Params | Notes |
|---|---|---|---|---|
| `cardiffnlp/twitter-roberta-base-sentiment-latest` | 2.4M | CC-BY-4.0 | unknown | Most popular; RoBERTa trained on tweets |
| `lxyuan/distilbert-base-multilingual-cased-sentiments-student` | 795K | Apache-2.0 | 135M (verified) | Multilingual DistilBERT; the only one with a verified size |
| `cardiffnlp/twitter-roberta-base-sentiment` | 253K | not listed | unknown | Older Cardiff NLP version |
| `finiteautomata/bertweet-base-sentiment-analysis` | 216K | not listed | unknown | BERTweet, tweet-oriented |
| `ahmedrachid/FinancialBERT-Sentiment-Analysis` | 164K | not listed | unknown | Finance-domain model |



**Hugging Face datasets**

| Dataset | Downloads | License | Notes |
|---|---|---|---|
| `mteb/multilingual-sentiment-classification` | 1.2K | unknown | Multilingual |
| `ai-forever/kinopoisk-sentiment-classification` | 573 | MIT | Russian movie reviews |
| `vikrantburman/fiqa-sentiment-classification` | 221 | MIT | Financial (FiQA) |
| `ArtGarfunkel/fiqa-sentiment-classification` | 143 | MIT | Financial (FiQA) |
| `ganchengguang/Text-Sentiment-Classification-and-Part-of-Speech-Sentiment-Classification-Mix-Dataset` | 111 | CC-BY-NC-4.0 | Mixed sentiment and part-of-speech labels; non-commercial |

The search doesn’t confirm that a dataset’s labels match a model’s, so check that before evaluating. The FiQA datasets pair naturally with `FinancialBERT`. None of these results is a typical English tweet or review benchmark, such as `tweet_eval` or `imdb`. I can search for those if you want a more standard comparison.

**Next step**

Tell me your text domain (tweets, reviews, finance, or something else). I can then inspect the candidate models and datasets, build a plan of up to 5 models on up to 1,000 rows, and run it locally on your CPU.

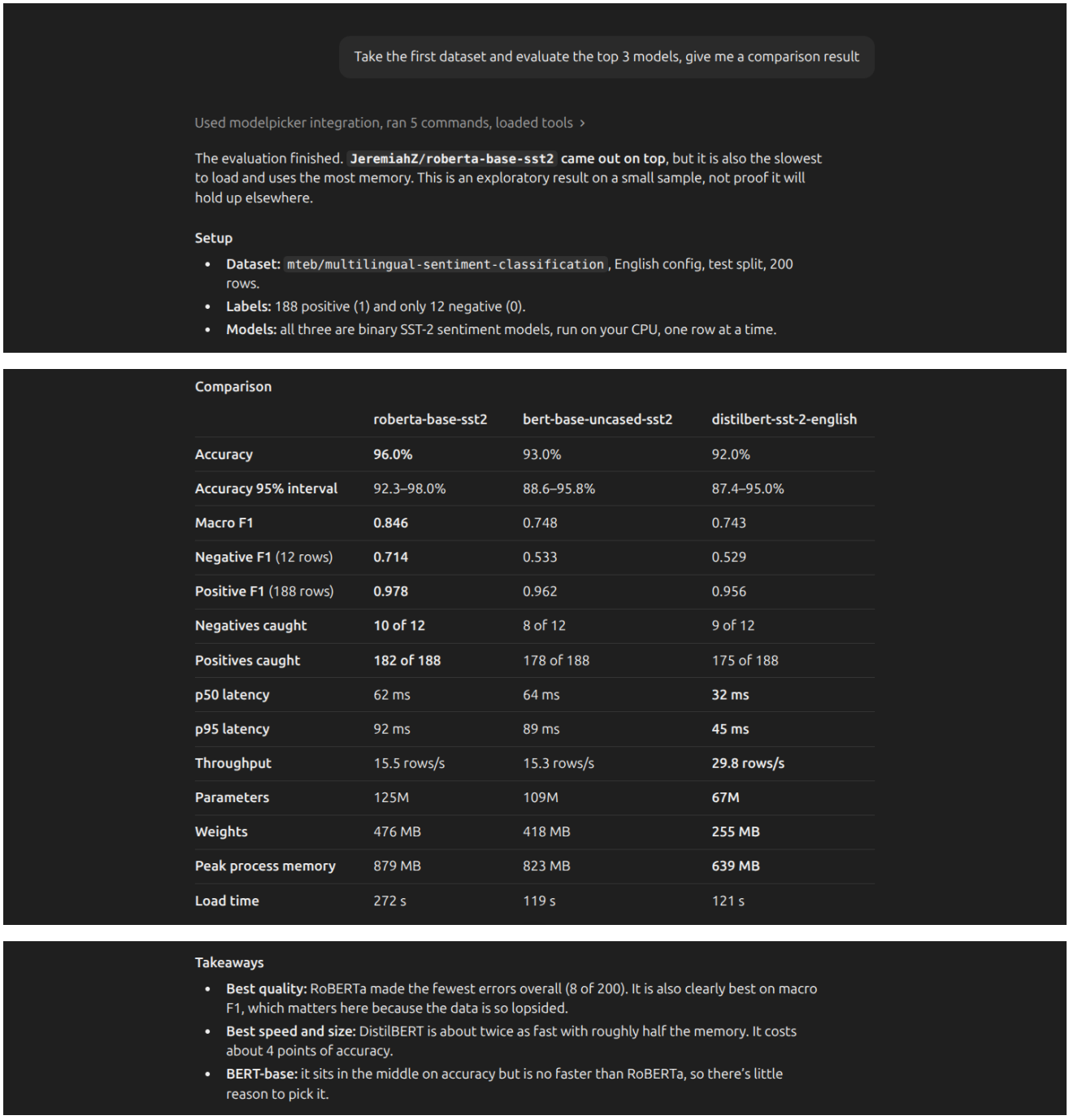
These are discovery results, not an evaluated ranking. Download counts, licenses and model metadata reflect the recorded session; compatibility requires inspection before evaluation.

# 03 / Evaluate & compare

### “Take the first dataset and evaluate the top 3 models, give me a comparison result”

Implemented workflow tools: inspect_huggingface_dataset · import_huggingface_dataset · inspect_model · create_evaluation_plan · start_evaluation · get_evaluation

View conversation




Verified against the saved evaluation: 200 English test rows (188 positive, 12 negative), 600 predictions, no execution errors. Accuracy: RoBERTa 96%, BERT 93%, DistilBERT 92%. Imbalanced, exploratory sample. Inference latency/throughput exclude download, loading and warmup; memory and weight figures use MiB.